# Steven Hoekstra

PhD Candidate in Economics, University of Amsterdam & Tinbergen Institute
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## Profile

I study how heterogeneous households and firms, boundedly rational expectations and market interactions shape monetary and fiscal policy transmission. I connect heterogeneous-agent equilibrium models and agent-based models, using neural-network solvers, simulation-based inference and microdata to assess economic mechanisms and the reliability of policy predictions.

## Research Interests

Macroeconomics, heterogeneous-agent models, agent-based modelling, expectations and bounded rationality, policy evaluation, networks, simulation-based inference, high-performance computing.

## Working Papers and Research Software

**Forecasting with a Dutch Agent-Based Model and Optimally Behaving Households**
Working paper, with Cars Hommes and Frank Pijpers. PDF
Runner-up, Society for Computational Economics Graduate Student Paper Award, CEF 2026

**When Do Estimated Macroeconomic Models Misread Wage–Price Transmission?**
Working paper; single-author. Identification, estimation and misspecification in an ABM estimated with neural simulation-based inference, and their consequences for forecasts and policy; known-truth simulation study and application to the 2022 Dutch inflation episode.

**BeforeIT.jl: High-Performance Agent-Based Macroeconomics Made Easy**
Research software; package co-author. Developed the validation suite, the CANVAS implementation and the multicountry forecasting pipeline.
Package · Original preprint (2025)

## Work in Progress

**Shortage or Misallocation? The Welfare Loss from Dutch Housing Policy, 2015–2035**
Single-author. Long-run welfare loss from housing shortage and mismatch under current Dutch policy, relative to first- and second-best benchmarks, across four levers: mortgage-interest subsidies, rent control, and nitrogen and grid-congestion constraints on construction. Spatial macroeconomic ABM with housing, labour mobility, demography and firm dynamics; calibration and welfare comparisons are ongoing.

**When Does Microdata Help? Conditioning Information in Estimated Structural Models**
Single-author. Early stage.

## Work Experience

**De Nederlandsche Bank — PhD Intern** Apr. — Dec. 2025
- Implemented the Bank of Canada CANVAS agent-based model for the Netherlands
- Estimated inflation channels using neural ratio estimation
- Wrote a note on the wage-price channel using this estimated model

**Statistics Netherlands (CBS) — Thesis Intern** Mar. — Sept. 2023
- Calibrated the agent-based model of Poledna et al. (2023) with Dutch register and national accounts data
- Developed an extension incorporating optimal consumption
- Ran simulations with both models and analysed their internal dynamics

**SEO Amsterdam Economics — Research Assistant** 2020 — 2021
- Researched academic pressure among secondary school students
- Conducted stakeholder interviews and set up a nationwide survey
- Performed statistical analysis on the survey outcomes

## Education

**PhD in Economics** — University of Amsterdam 2023 — current
Supervisors: Prof. dr. C.H. Hommes and Prof. dr. F.P. Pijpers
Expected thesis submission: May 2027; expected defence: September 2027

**MPhil Economics and Econometrics** — Tinbergen Institute ***(cum laude)*** 2021 — 2023
Specialisation: Complexity, Mathematics, and Computation
Thesis: An Agent Based Model of the Dutch Economy
Supervisors: Prof. dr. C.H. Hommes and Prof. dr. F.P. Pijpers

**MSc Econometrics** — University of Amsterdam ***(cum laude)*** 2020 — 2021
Specialisation: Complexity and Economic Behaviour
Thesis: Endogenous Financial Instability via Heterogeneous Beliefs
Supervisor: Prof. dr. C.H. Hommes

**MSc Economics** — London School of Economics 2019 — 2020
Specialisation: Public Economics
Thesis: The Inequality of Education using a Two Agent New Keynesian Model
Supervisor: Prof. dr. F. Cowell

**Exchange Programme** — Royal Melbourne Institute of Technology 2018 — 2019
Minor in Finance and Programming

**BSc Economics and Business Economics** — University of Groningen 2016 — 2019
Thesis: The Impact of News Coverage on Security Prices

**BSc Econometrics** — University of Groningen 2015 — 2018
Thesis: Arbitrage Smoothing and Term Structure Models

## Grants

**NWO Compute Grants — Snellius**
Three awarded compute grants of 1,000,000 CPU hours each on Snellius, the Dutch national supercomputer; 3,000,000 CPU hours in total.

## Teaching

### Graduate

**Behavioural Macro and Finance — Lecturer** 2025
Developed and taught the MSc lecture series with Cars Hommes, University of Amsterdam.

**MPhil guest lectures — Tinbergen Institute**
Annual guest lectures in Behavioural Macroeconomics and Complexity.

### Undergraduate

**BSc Econometrics thesis supervision — University of Amsterdam**
Ongoing.

**Mathematics 3: Advanced Linear Algebra — Tutor** 2025 — 2026

**Mathematics 2: Linear Algebra — Tutor** 2024 — 2025

**Microeconomics for AE — Tutor** 2023 — 2024

## Presentations

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| 2026 | Computing in Economics and Finance (CEF), Society for Computational Economics |
| 2026 | Workshop on Economic Science with Heterogeneous Interacting Agents (WEHIA) |
| 2025 | Agent-Based Modelling for Policy (ABM4Policy), European Central Bank, Frankfurt |
| 2025 | 7th Behavioral Macroeconomics Workshop, University of Bamberg |
| 2025 | MacroABM 2nd Workshop, *Microdata in Macromodels*, Vienna University of Economics and Business |
| 2025 | Workshop on Economic Science with Heterogeneous Interacting Agents (WEHIA) |
| 2025 | De Nederlandsche Bank, Amsterdam |
| 2023 | De Nederlandsche Bank, Amsterdam |

## Honours and Scholarships

### Honours

**Runner-up — Society for Computational Economics Graduate Student Paper Award** 2026
Computing in Economics and Finance (CEF), for Forecasting with a Dutch Agent-Based Model and Optimally Behaving Households.

**Cum Laude — MPhil Economics and Econometrics, Tinbergen Institute** 2023

**Cum Laude — MSc Econometrics, University of Amsterdam** 2021

### Scholarships

**Full scholarship (merit based) — Tinbergen Institute** 2022

**Tuition fee waiver — Tinbergen Institute** 2021 — 2023

## Visits and Summer Schools

**Visiting Scholar — IIASA, Laxenburg** Summer 2022
International Institute for Applied Systems Analysis. Visited Dr. S. Poledna to work on agent-based models

**Tools for Macroeconomists: Advanced Tools — University of Oxford** Aug. 2021
Doctoral-level course on DSGE models: solving and simulating models with heterogeneous agents, continuous time models, occasionally binding constraints and bounded rationality

**Tools for Macroeconomists: The Essentials — University of Oxford** Aug. 2021
Doctoral-level course on DSGE models: numerical analysis, perturbation and projection methods, Blanchard-Kahn conditions and sunspot solutions, the Kalman filter and Bayesian estimation

## Skills

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| Programming | Matlab, Python, R, Julia, C, SQL, Bash |
| High-performance computing | Parallel computing (MPI, BSPlib, Slurm) |
| Methods | Agent-based modelling, heterogeneous-agent models, neural-network solvers, simulation-based inference, network methods, machine learning, econometrics, dynamic stochastic general equilibrium modelling (Dynare) |

## References

**Prof. dr. C.H. Hommes**
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**Prof. dr. F.P. Pijpers**
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